

SCIENTIFIC RESEARCH METHODS

Software manual

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Science Research Methods: Software

An introduction to quantitative research and statistics: Using software

Contents

Preface	v
1 Introduction	1
I Using jamovi	3
2 jamovi: general information	5
3 jamovi: inference for two odds or two proportions	7
3.1 Introduction	7
3.2 Data entry	8
3.3 Analysis	9
3.4 Numerical summaries	10
4 jamovi: inference about the difference between two means	13
4.1 Introduction	13
4.2 Data entry	13
4.3 Analysis	15
4.4 Numerical summaries	16
5 jamovi: graphs	17
5.1 Bar charts (side-by-side; stacked)	17
5.2 Boxplots	18
5.3 Error bar charts	19
5.4 Editing graphs	19
II Using SPSS	21
6 SPSS: general information	23
7 SPSS: inference for two odds or two proportions	25
7.1 Introduction	25
7.2 Data entry	26
7.3 Analysis	26
7.4 Numerical summaries	27

8	SPSS: inference about the difference between two means	29
8.1	Introduction	29
8.2	Data entry	29
8.3	Analysis	30
8.4	Numerical summaries	32
9	SPSS: graphs	33
9.1	Side-by-side bar charts	35
9.2	Stacked bar charts	36
9.3	Boxplots	37
9.4	Error bar charts	38
9.5	Editing graphs	39
10	Data files	43

Preface

This book has been prepared for use with the book *Scientific Research and Methodology*, which is an introduction to quantitative research methods in the scientific, engineering and health disciplines.

This book introduces the whole research process, from asking a research question to analysis and reporting of the data. The focus, however, is on the analysis of data.

Statistical software

This manual explains how to conduct some of the analyses in the book *Scientific Research Method*¹ using software. We discuss using jamovi² ([The jamovi Project, n.d.](https://www.jamovi.org/)) and SPSS³ ([IBM Corp 2016](https://www.ibm.com/products/spss-statistics)) only.

jamovi is *free* and is like (but not exactly the same as) SPSS. From the jamovi homepage (sic)⁴:

jamovi is a new ‘3rd generation’ statistical spreadsheet. designed from the ground up to be easy to use, jamovi is a compelling alternative to costly statistical products such as SPSS and SAS.

Callouts used on this book

The following callouts are used in this book:



These chunks introduce the objectives for the chapters of the book.



These chunks offer helpful information.

¹<https://bookdown.org/pkaldunn/SRM-Textbook/>

²<https://www.jamovi.org/>

³<https://www.ibm.com/products/spss-statistics>

⁴<https://www.jamovi.org/>



These chunks highlight common mistakes or warnings, about a particular concept or about using a formula.



These chunks point to the data files being used, collated in Sect. 10, so you can download the data, and follow the instructions.

Using statistical software rather than a spreadsheet

(The following information is based on (Dunn 2025).)

Many people use spreadsheets (such as Microsoft Excel) rather than statistical software. But using spreadsheets requires extreme care; many extremely expensive and dangerous errors have been made through using spreadsheets (AlTarawneh and Thorne 2017), including problems when reporting the 2020 COVID-19 pandemic⁵.

These problems emerge for different reasons:

- Spreadsheets often change the entered data (for example, reformatting entries as dates if the spreadsheet *thinks* the data should be a date) in the interest of being ‘user-friendly’, even when not appropriate. This has had dire consequences (Ziemann, Eren, and El-Osta 2016).
- Many spreadsheets have errors in formulas (R. R. Panko and Sprague Jr 1998), but these are incredible difficult to locate and hence fix (Galletta et al. (1996), R. Panko (2016)).
- Spreadsheets do not leave a record of how the data have been analysed; for example, formulas can be very difficult to understand and parse. Keeping a record of the analysis, new variables that have been created, and other operations with the data is called **reproducible research**. Reproducibility ensures, among other advantages, that the results can be checked by others.
- Excel has bugs (Keeling and Pavur (2004), Mélard (2014)) even in very basic operations (Berger (2007), Hargreaves and McWilliams (2010)). After trying to fix these bugs, sometimes they are made even worse (McCullough and Wilson 2002).

Spreadsheets can be used for research and analysis... but you must be very careful! Many of the problems are due to human error, and some emerge because Excel is being used for purposes it is not really designed for (i.e. scientific analysis).

Using jamovi and SPSS

jamovi

We **strongly recommend** installing the most recent **stable, desktop** (not online) version of jamovi⁶.

⁵<https://www.zdnet.com/article/excel-errors-microsofts-spreadsheet-may-be-hazardous-to-your-health/>

⁶<https://www.jamovi.org/download.html>

SPSS

All recent versions of SPSS work in similar ways. SPSS is installed in computer labs and available using *USC Anywhere*⁷.

Learning Outcomes

In this book, you will learn to:

- Use jamovi to conduct chi-square tests.
- Use jamovi to conduct two-sample t -tests.
- Use jamovi to create graphs.
- Use SPSS to conduct chi-square tests.
- Use SPSS to conduct two-sample t -tests.
- Use SPSS to create graphs.

How to cite this book

Peter K. Dunn (2019). *Scientific Research Methods: Software manual*. <https://bookdown.org/pkaldunn/SRM-software/>

⁷<https://anywhere.usc.edu.au/vpn/index.html>

1

Introduction

Before you start to use jamovi or SPSS, you need to know *why* you need the software.

- If your RQ concerns the **relationship between two qualitative variables**:
 - Analysis of **odds ratios** using, for example, a χ^2 -test; or
 - Analysis of **two proportions** using, for example, a z -test.
 - Displays using a side-by-side or stacked barchart.
- If your RQ concerns the **difference between means in two groups**:
 - Analysis of two means using, for example, a two-sample t -test.
 - Displays using a error bar chart (and perhaps a boxplot).

The data files used in this document are available in Sect. 10.

You can download the data, and follow the instructions in this document.

Part I

Using jamovi

2

jamovi: general information



Topics covered in this chapter:

- Tips for working with jamovi.

Before starting analysis with jamovi¹ ([The jamovi Project, n.d.](https://www.jamovi.org/)), here are some **general tips**:

- jamovi is a very new package (built on R ([R Core Team 2025](https://www.r-project.org/))), and currently lacks some desirable features. These will no doubt arrive in due time.
- jamovi is **free** to download, from the jamovi webpage².
- *Except for graphs*, jamovi output should **not** be presented in reports or presentations. Instead, the information from jamovi output should be used to construct proper tables, or to produce proper conclusions.
- The information about the variables must be correctly set up **before** analysis.
- In jamovi (as with most statistical software), *each row represents one unit of analysis*.
- In jamovi (as with most statistical software), *each column represents one variable*.
- At the top of each column (but **not** in Row 1!) is the *name* of the variable; *double click* on the variable name and set information about the variables there (Fig. 2.1). Then you can tell jamovi information about the variables, such as whether the data are nominal, ordinal or ‘continuous’ (which is what jamovi calls *all* quantitative variables...), and the *levels* of the nominal and ordinal variables (Fig. 2.2).
- You can get help with using jamovi by:
 - Watching software screencasts on Canvas.
 - Using jamovi resources³.
 - Using LinkedIn Learning⁴.
 - Watching videos⁵.
 - Asking SCI110 staff **in tutorials**, or in the Canvas *Discussions*.

¹<https://www.jamovi.org/>

²<https://www.jamovi.org/>

³<https://www.jamovi.org/user-manual.html>

⁴<https://www.linkedin.com/learning/>

⁵<https://datalab.cc/tools/jamovi>

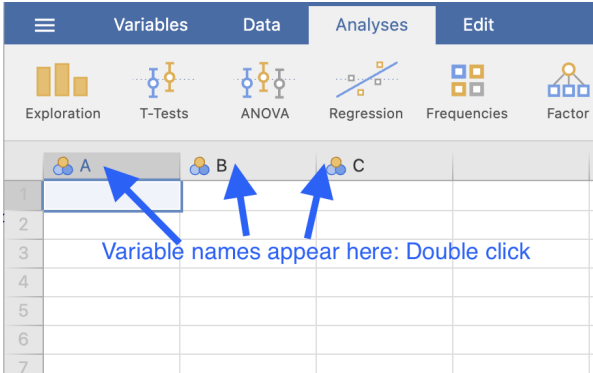


FIGURE 2.1: Click on the column headers to describe the variables

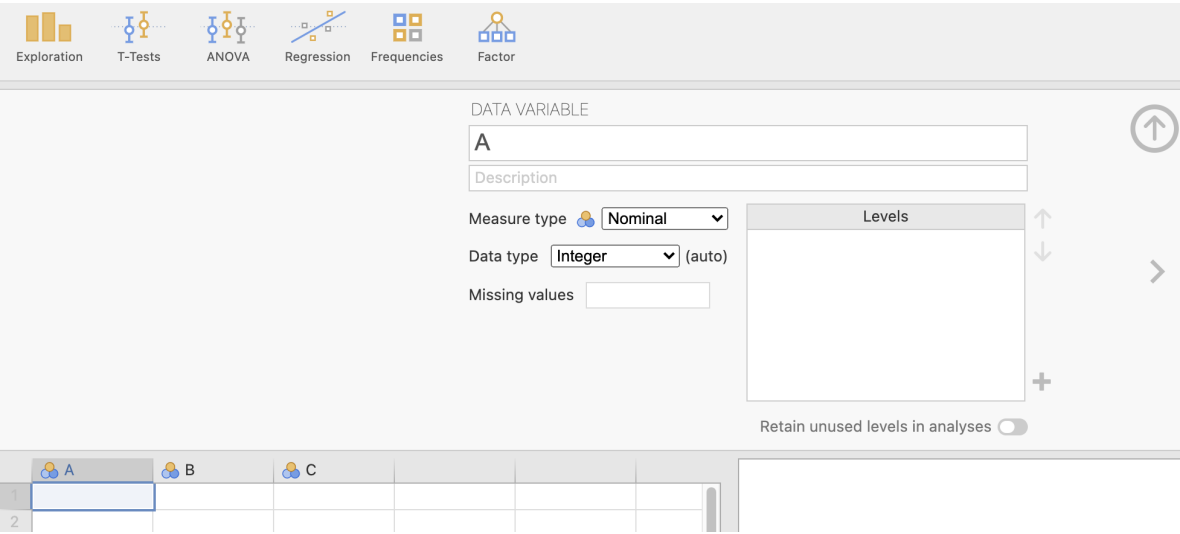


FIGURE 2.2: The information to fill in for each variable

3

jamovi: inference for two odds or two proportions



Topics covered in this chapter:

- How to use jamovi for comparing two odds (i.e., analysis using odds ratios).
- How to use jamovi for comparing two proportions.

3.1 Introduction

RQs comparing *two qualitative variables* require count data that can be compiled into a two-way table of counts. Consider data (Gammon et al. 2012) in Exercise 31.16 of the textbook¹ (Table 3.1).



The data file (B12-Long.csv) can be downloaded in Sect. 10. You can download the data, and follow the instructions.

This data set is available from the textbook as B12-Long in Appendix A². You can download the data and follow along if you wish.

There are *two qualitative variables* being measured on each individual:

1. Whether the person is vegetarian (or not);
2. Whether the person is B12 deficient (or not).

Hence, jamovi will have *two columns*: one for each variable. There are 124 units of analysis, so there are 124 *rows* of data.

TABLE 3.1: The number of vegetarian and non-vegetarian women who are (and are not) B12 deficient

	B12 deficient	Not B12 deficient	Total
Vegetarians	8	26	34
Non-vegetarians	8	82	90
Total	16	108	124

¹<https://bookdown.org/pkaldunn/SRM-Textbook/AnalysisOddsRatio.html>

²<https://bookdown.org/pkaldunn/SRM-Textbook/AppendixDataSets.html>

3.2 Data entry

In jamovi, the two columns are set up as follows:

- **Diet:** 1 means ‘Vegetarian’ (i.e. in Row 1 of the data table), and 2 means ‘Non-vegetarian’ (i.e. in Row 2 of the data table in Table 3.1).
- **B12:** 1 means ‘B12 deficient’ (i.e. in Column 1 of the data table), and 2 means ‘Not B12 deficient’ (i.e. in Column 2 of the data table in Table 3.1); see Fig. 3.1.

The data needs to be set up so that *each row represents one unit of analysis*. That means there will be 124 rows of data: one for each person (unit of analysis).

In jamovi, the data are entered as shown in Fig. 3.1; there are 124 rows, but we only show some of these rows to save space.



Each *row* in the jamovi data represents one *unit of analysis*.

Each *column* in the jamovi data represents one *variable*.

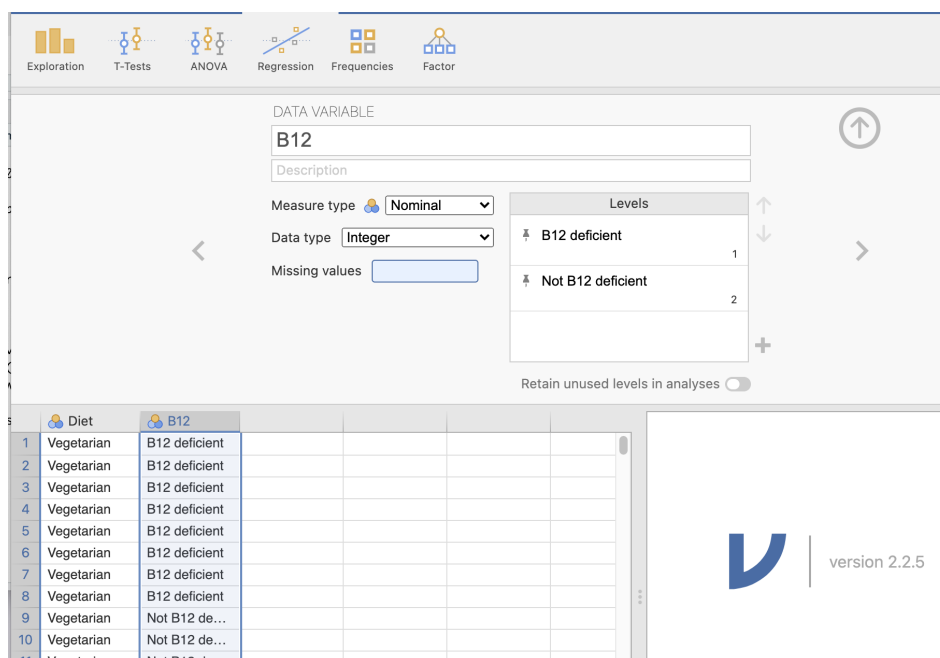


FIGURE 3.1: The B12 data entered into jamovi



Note that I entered 1 and 2 into the jamovi worksheet: jamovi just *shows* what these mean in the worksheet, because I have told jamovi what the 1 and the 2 mean.

Your data is now set up and ready to use.

3.3 Analysis

Remember that **this data set is available from the textbook as B12–Long** in Appendix A³

Once the data are entered, we can begin analysis:

1. **Use the menu:** select **Analyses> Frequencies> Independent samples (χ^2 test of association)**

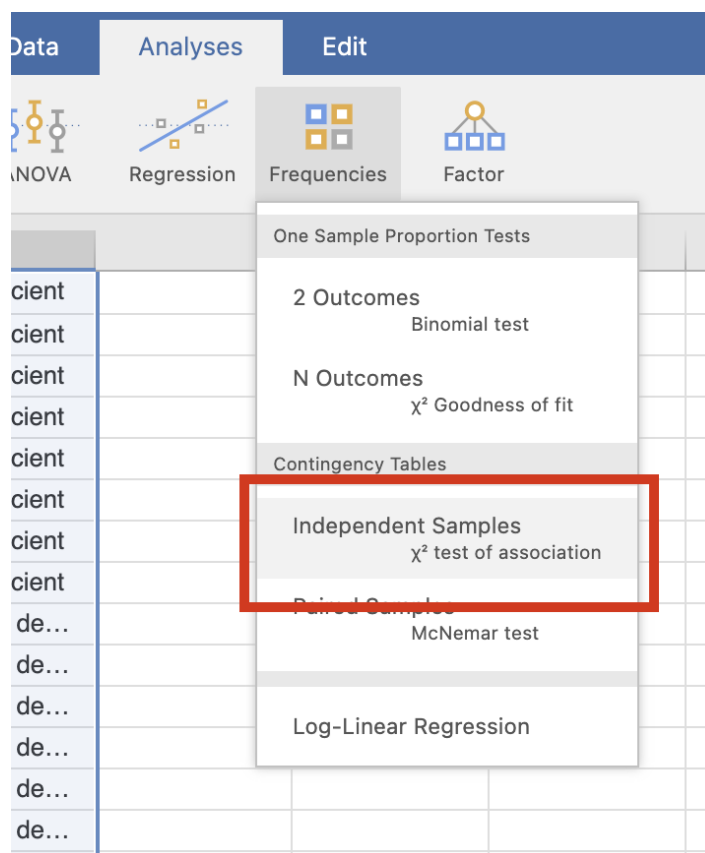


FIGURE 3.2: Select the correct test

2. **Select the variables:** dragging them from the left panel to the **Rows** and **Columns** areas as appropriate. In Table 3.1, **Diet** is in the rows of the table, for example.
3. **Select the analysis options:** Under the area where you entered the variables, select the right-pointing arrow beside **Statistics**, and then:
 - Select χ^2 : this conducts the χ^2 hypothesis test for comparing odds.
 - Select z test for difference in 2 proportions: this conducts the z-test to compare proportions.
 - Select Odds ratio: this shows the value of the odds ratio.

³<https://bookdown.org/pkaldunn/SRM-Textbook/AppendixDataSets.html>

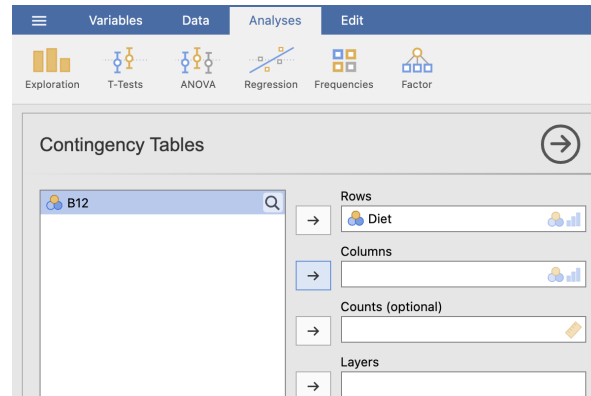


FIGURE 3.3: Enter the variables

- Select `Difference in proportions`: this shows the difference between the two proportions.
- Select `Confidence intervals`: this shows the CI for the odds ratio, and for the difference between the proportions.

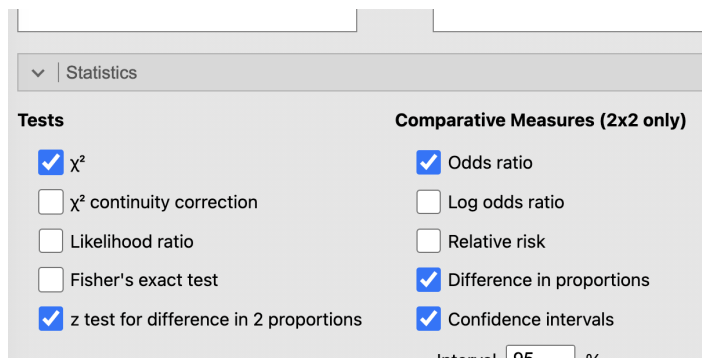


FIGURE 3.4: Check the correct options

4. You will have your output in the jamovi Output window on the right.

3.4 Numerical summaries

Percentages can be obtained from the same place: **Analyses > Frequencies Independent samples (χ^2 test of association)**. Under the area where you selected the **Statistics**, select the right-pointing arrow beside **Cells**, and then:

- Select `Row` under the **Percentages** section; and/or
- Select `Column` under the **Percentages** section.

Of course, you can select any of the other options too, and see what happens.

The figure shows a software interface for selecting what to show in a table. The interface has a sidebar on the left with the following options:

- > | Statistics
- ▼ | Cells
- > | Plots

The 'Cells' section is expanded, showing two sub-sections:

- Counts**
 - ☒ Observed counts
 - ☐ Expected counts
- Percentages**
 - ☒ Row
 - ☐ Column
 - ☐ Total

FIGURE 3.5: Select what to show in the table

4

jamovi: inference about the difference between two means



Topics covered in this tutorial:

- Learn how to use jamovi for comparing two means.

4.1 Introduction

Two independent sample t -tests are used to compare the means of two separate groups.

Consider the face-plant data (Wojcik et al. 1999) used in Exercise 30.11 of the textbook¹ (Table 4.1).



The data file (`forwardfall.csv`) can be downloaded in Sect. 10.

You can download the data, and follow the instructions.

TABLE 4.1: *Lean-forward angles for older and younger women*

Younger women		Older women
29	34	18
32	27	15
34	32	23
31	28	13
33	27	12

4.2 Data entry

In the **Data** area, each row represents a unit of analysis (in this case, a person), as is usual in jamovi (Fig. 4.1).

Each unit of analysis gets two variables recorded: one is quantitative (**Lean angle (in degrees)**), and one is qualitative (**Age group**). So there are two columns in the data, one for each variable.

¹<https://bookdown.org/pkaldunn/SRM-Textbook/AnalysisTwoMeans.html>

The order of the variables doesn't matter, but here the quantitative variable is listed first (the lean-forward angle), and then the qualitative variable (the age group).

The variables need to be properly set up (by double-clicking on the variable names); see Figs. 4.2 and 4.3.

	Exploration	T-Tests	ANOVA
	LeanAngle	Group	
1	29	Younger fe...	
2	34	Younger fe...	
3	33	Younger fe...	
4	27	Younger fe...	
5	28	Younger fe...	
6	32	Younger fe...	
7	31	Younger fe...	
8	34	Younger fe...	
9	32	Younger fe...	
10	27	Younger fe...	
11	18	Older females	
12	15	Older females	
13	23	Older females	
14	13	Older females	
15	12	Older females	

FIGURE 4.1: The data entered in jamovi

The screenshot shows the Jamovi software interface. At the top, there are tabs for different statistical tests: Exploration, T-Tests, ANOVA, Regression, Frequencies, and Factor. Below these tabs, the 'DATA VARIABLE' section is active, showing the variable 'LeanAngle'. The description for 'LeanAngle' is 'Maximum lean angle (in degrees)'. The 'Measure type' is set to 'Continuous' and the 'Data type' is set to 'Decimal'. There is a 'Missing values' field which is currently empty. To the right of these settings is a 'Levels' section, which is currently empty. At the bottom of the interface, a preview of the data table is visible, showing the first seven rows of the data entered in Figure 4.1.

	LeanAngle	Group
1	29	Younger fe...
2	34	Younger fe...
3	33	Younger fe...
4	27	Younger fe...
5	28	Younger fe...
6	32	Younger fe...
7	31	Younger fe...

FIGURE 4.2: Describing the quantitative variable

The two age groups are defined so that a 1 represents younger women, and a 2 represents older women. In the jamovi worksheet, the values 1 and 2 were entered, but jamovi shows what these mean in the worksheet when these values are defined.

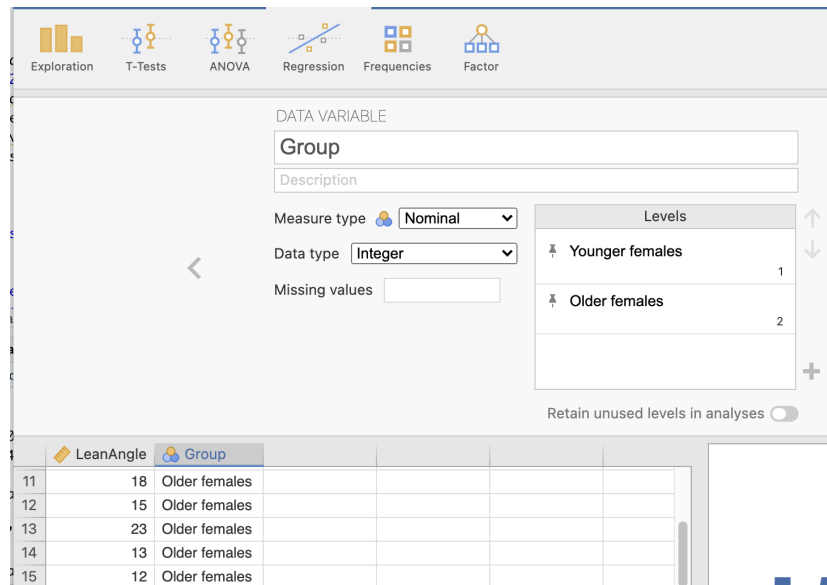


FIGURE 4.3: Describing the qualitative variable

4.3 Analysis

Remember, **this data set is available from the textbook as `forwardfall.csv` in Appendix A²**. You can download the data and follow along if you wish.

1. **Use the jamovi menu:** select **Analyses> T-Tests> Independent Samples T-Test**
2. **Select the Variables.**

- For the **Dependent Variables**, add the quantitative variable (here, `LeanAngle`).
- For the **Grouping Variable**, add the qualitative variable (here, `Group`).

3. Then select the options:

- `Student's and Welch's` should be select to perform the hypothesis test and compute the CIs.
- `Mean difference` to show the difference between the two means in the output.
- `Confidence interval` to show the CI for the difference between the two means.
- `Descriptives` to show the numerical summary of each group.
- `Descriptives plot` to show the error bar chart.



Even if you have a one-tailed hypothesis, keep the hypotheses as the two-tailed (first) option, or you will get some strange output. The one-tailed P -value will be half the two-tailed P -value.

4. You will have your output in the jamovi Output window.

²<https://bookdown.org/pkaldunn/SRM-Textbook/AppendixDataSets.html>

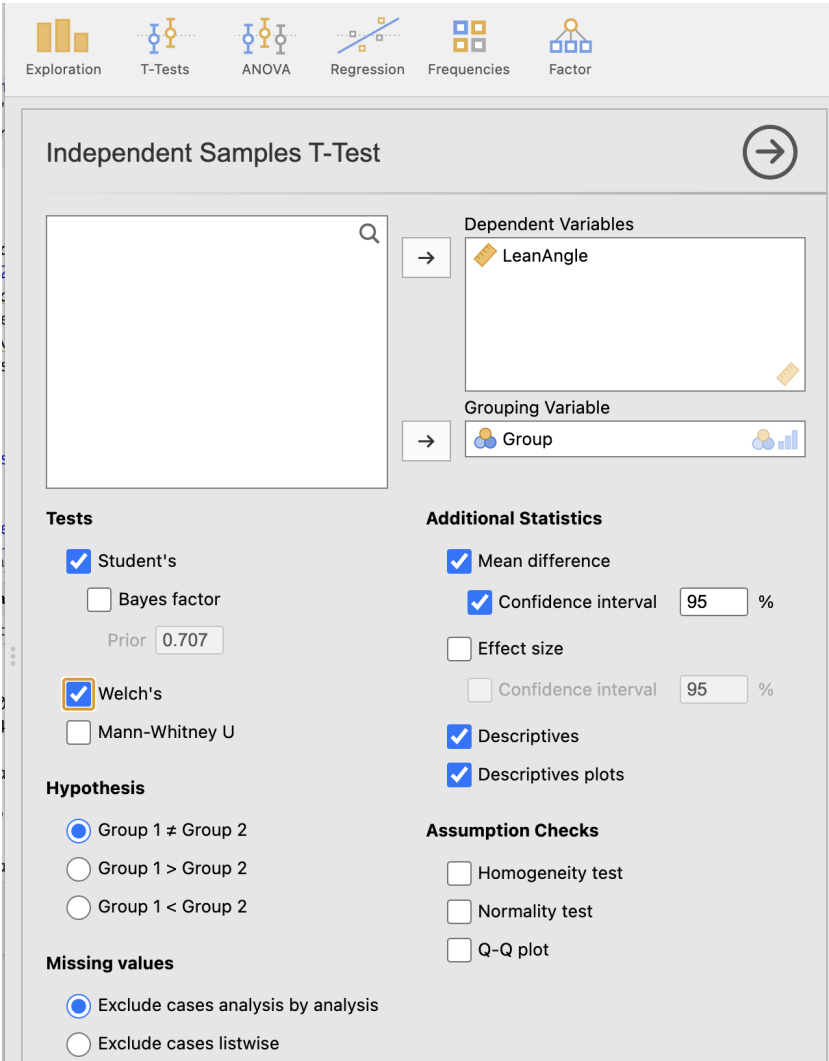


FIGURE 4.4: Enter the variables

4.4 Numerical summaries

The *mean*, *median*, *standard deviation* and *standard error* of each group are produced as part of the output from the *t*-test analysis step above.

The *mean* and *standard error* for the *difference* is obtained from the *t*-test output too.

5

jamovi: graphs



Topics covered in this tutorial:

- Creating graphs in jamovi.



The data files used in this section can be downloaded in Sect. 10.

You can download the data, and follow the instructions.

Graphs in jamovi are sometimes produced as part of the **Analyses**, and sometimes through the **Exploration** menu.

5.1 Bar charts (side-by-side; stacked)

For this section, the data introduced in Sect. 3.1 are used.

1. **Use the jamvoi menu:** select **Analyses> Frequencies> Independent samples (χ^2 test of association)**.
2. **Add the variables.** Add the variables to the rows and columns (in this case, the Rows is Diet and the Columns is B12).
3. Click on the right-pointing arrow next to the **Plots** option from the bottom of the left-side screen.
4. In the **Plots** section, select **Bar Plot**; then select **Bar Type** (either a **Side by side bar chart**, or a **Stacked bar chart**).
5. You can change the vertical axis to represent either counts or percentages by your selection under **Y-Axis**.
6. You can change which variables is on the horizontal axis by your selection under **X-Axis**. There are plenty of options, so feel free to experiment. A useful one is to use for the **Y-Axis** is **Percentages with the within rows option**.
6. You will have your graph in the Output window.

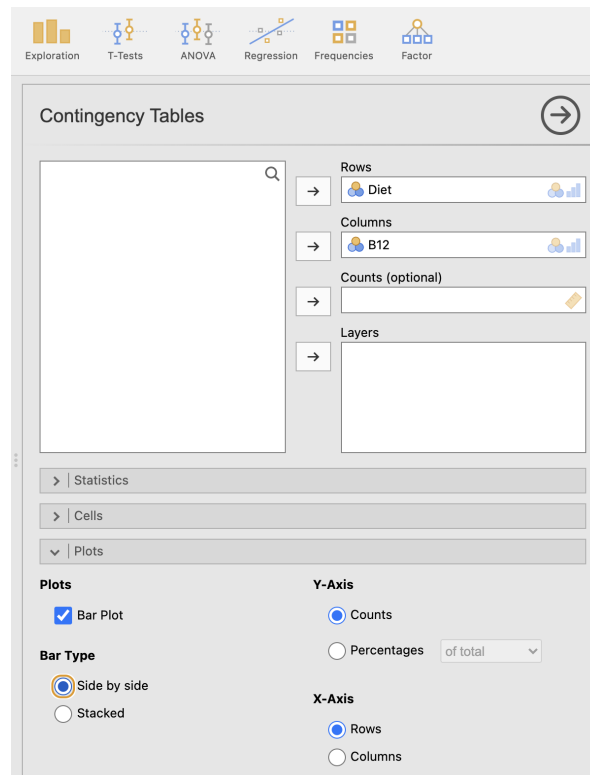


FIGURE 5.1: Adding the variables to create a side-by-side bar chart in jamovi

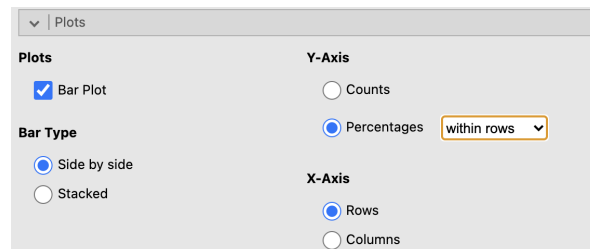


FIGURE 5.2: Adding the variables to create a side-by-side bar chart in jamovi

5.2 Boxplots

For this section, the data introduced in Sect. 4.1 are used.

1. **Use the jamovi menu:** select **Analyses> Exploration> Descriptives**.
2. **Add the variables.** Place the quantitative variable in the *Variables* area, and the qualitative variable in the **Split by** area.
3. Then, click on the right-pointing arrow next to **Plots**, and in the **Box Plots** area select **Box plot**.
4. You will have your graph in the jamovi Output window.

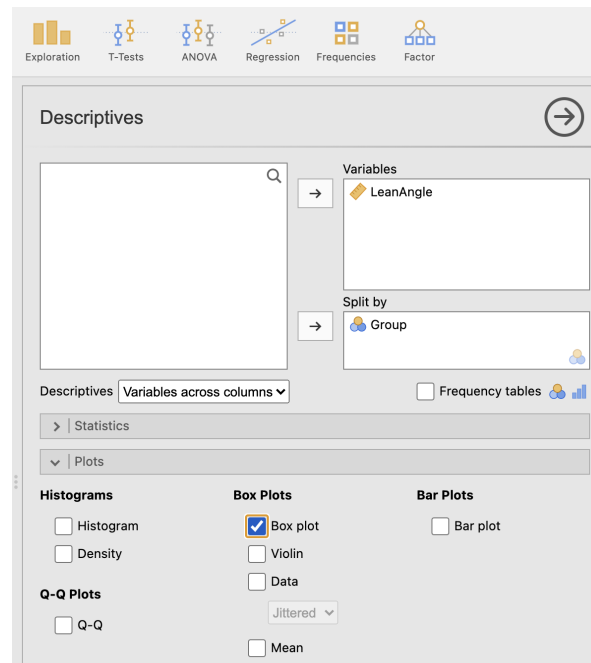


FIGURE 5.3: *Selecting the variables for a boxplot*

5.3 Error bar charts

Error bar charts are produced as part of the *analysis*: see Chap. 4.

5.4 Editing graphs

At the time these notes were prepared, *jamovi does not allow graphs to be edited* (but it is promised to be coming soon!).

Part II

Using SPSS

6

SPSS: general information



Topics covered in this chapter:

- Tips for working with SPSS.

Before starting analysis with SPSS (IBM Corp 2016), here are some **general tips**:

- In general, you cannot download a copy of SPSS for use at home (which you *can* do with jamovi).
- *Except for graphs*, SPSS output should **not** be presented in reports or presentations. Instead, the information from SPSS output should be used to construct proper tables, or to produce proper conclusions.
- SPSS has two worksheets sheets; **both** must be set up correctly (you can switch between the two worksheets by clicking on these at the bottom of the SPSS window):
 - The **Data View**, which contains the data.
 - The **Variable View**, which contains information *about* each variable.
- In SPSS (as with most statistical software), *each row represents one unit of analysis* in the **Data View**.
- In SPSS (as with most statistical software), *each column represents one variable* in the **Data View**.
- At the top of each column (but **not** in Row 1!) is the *name* of the variable;
- When setting up the **Variable View**, many columns can be filled in (Fig. 6.1). These are the important columns to get right (the others aren't important for our purposes):
 - **Name**: A short description of the variable (with no spaces or punctuation!); for example DiaBP.
 - **Type**: Usually **Numeric** (apart from *names* which can be **String**).
 - **Label**: A fuller description of the variable. This is what will appear in tables and graphs to describe the variable. For example Diastolic blood pressure (in mm Hg).
 - **Values**: For *qualitative* variables only: tell SPSS what each number represents (for example, does a 1 represent females, or males?)
 - **Measure**: *This is important*: You must tell SPSS whether each variable is nominal, ordinal, or scale (i.e. quantitative) by using the drop-down options.
- You can get help with using SPSS by:
 - Using LinkedIn Learning¹.
 - Watching software screencasts on Canvas.
 - Borrowing books in the library.
 - Asking SCI110 staff, including using the Canvas *Discussions*.

¹<https://www.linkedin.com/learning/>

Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role

FIGURE 6.1: Information that can be given for each variable in the Variable View window)

7

SPSS: inference for two odds or two proportions



Topics covered in this tutorial:

- How to use SPSS for comparing two odds (i.e., analysis using odds ratios).
- How to use SPSS for comparing two proportions.

7.1 Introduction

χ^2 tests ('*ki-squared tests*') are used to analyse data where each unit of analysis has *two qualitative variables* measured (or observed) about it. This type of data can be compiled into a two-way table of counts.

Consider data (Gammon et al. 2012) used in Exercise 31.16 of the textbook¹ (Table 7.1).

Two qualitative variables being measured on each individual:

1. Whether the person is vegetarian (or not);
2. Whether the person is B12 deficient (or not);

So SPSS will have *two columns*: one for each variable.



The data file (B12-Long.sav) can be downloaded in Sect. 10.

You can download the data, and follow the instructions.

TABLE 7.1: The number of vegetarian and non-vegetarian women who are (and are not) B12 deficient

	B12 deficient	Not B12 deficient	Total
Vegetarians	8	26	34
Non-vegetarians	8	82	90
Total	16	108	124

¹<https://bookdown.org/pkaldunn/SRM-Textbook/AnalysisOddsRatio.html>

7.2 Data entry

In the SPSS **Variable View**:

- **Diet**: 1 means ‘Vegetarian’ (i.e. in Row 1 of the data table), and 2 means ‘Non-vegetarian’ (i.e. in Row 2 of the data table).
- **B12**: 1 means ‘B12 deficient’ (i.e. in Column 1 of the data table), and 2 means ‘Not B12 deficient’ (i.e. in Column 2 of the data table).

See Fig. 6.1.

The usual approach for data entry used by SPSS is that *each row represents one unit of analysis*. That means that there will be 124 rows of data: one for each person (unit of analysis).

In SPSS, the data are entered as shown in Fig. 7.1, but there would be 124 rows.

Diet	B12
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	2
1	2
1	2
1	2

FIGURE 7.1: Data entered into SPSS

The **Variable View** would describe these two variables; see Fig. 7.2.

	Name	Type	Width	Deci...	Label	Values	Mis...	C...	Align	Measure	Role
1	Diet	Numeric	8	0	Diet	{1, Vegetarian}...	None	8	Ri...	Nominal	Input
2	B12	Numeric	8	0	B12 Deficiency	{1, B12 deficient}...	None	8	Ri...	Nominal	Input

FIGURE 7.2: Describing the variable in the Variable View window

7.3 Analysis

Once the data are entered, we can begin analysis:

1. **Use the menu:** select **Analyze> Descriptive Statistics> Crosstabs...**

2. **Select the variables**, dragging them from the left panel to the **Row(s)** and **Column(s)** areas as appropriate. In Table 7.1, **Diet** is in the rows of the table, for example. See Fig. 7.3.

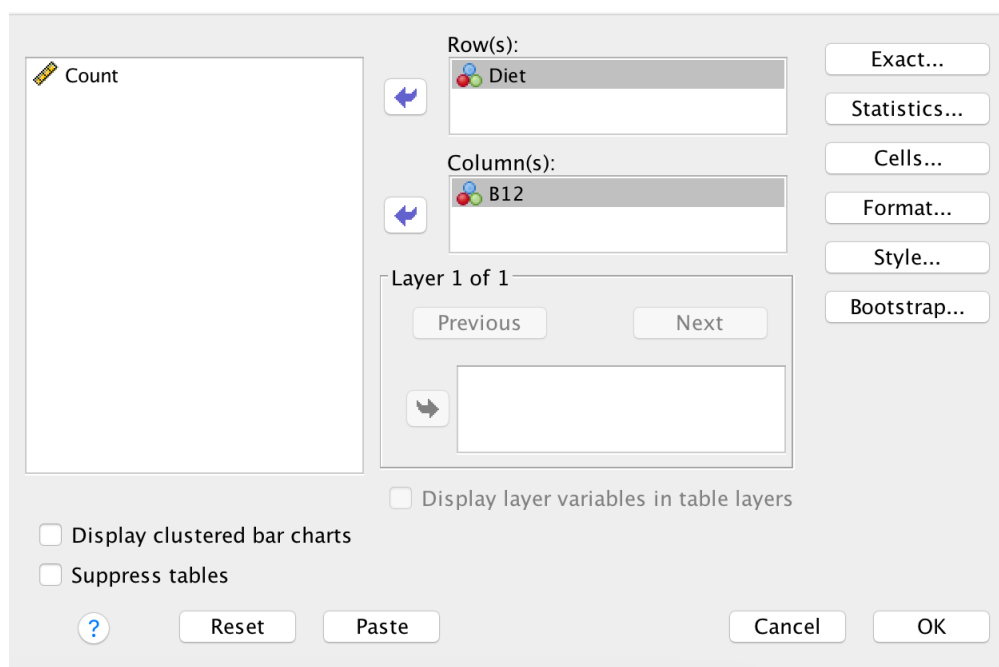


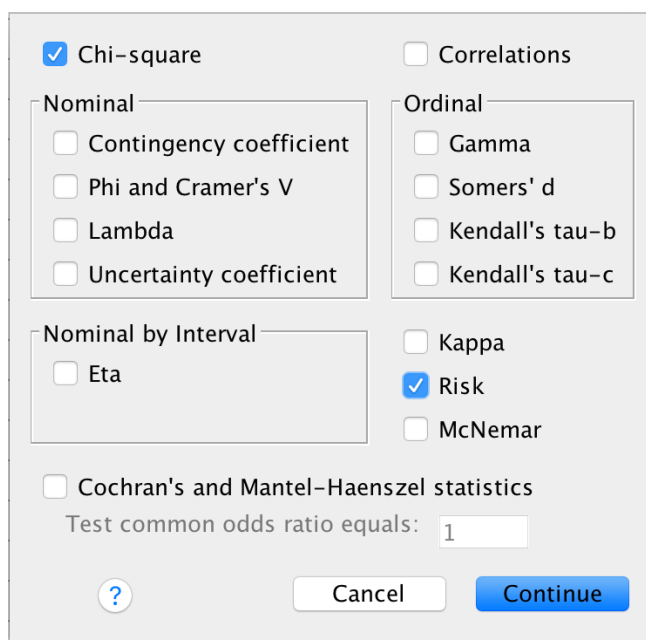
FIGURE 7.3: Add the variables

3. **Select the analysis options.** Click the **Statistics...** button, and then (Fig. 7.4):
 - Select the **Chi-square**: to produce information for the hypothesis test.
 - Select the **Risk** option: to produce a CI for the odds ratio.
4. Press **Continue**, and then press **OK**. You will have your output in the SPSS Output window.

7.4 Numerical summaries

Percentages can be obtained from **Analyze > Descriptive Statistics > Crosstabs...** In the window that appears, select **Cells...** and then select **Percentages** as **Row**, as **Column**, or even as both if you wish.

Odds are computed as part of the inference output from Sect. 7.3.



The image shows the 'Chi-square' options dialog box in SPSS. The 'Chi-square' checkbox is checked. Under the 'Nominal' section, 'Contingency coefficient', 'Phi and Cramer's V', 'Lambda', and 'Uncertainty coefficient' are all unchecked. Under the 'Ordinal' section, 'Gamma', 'Somers' d', 'Kendall's tau-b', and 'Kendall's tau-c' are all unchecked. Under the 'Nominal by Interval' section, 'Eta' is unchecked. In the middle right section, 'Kappa' and 'McNemar' are unchecked, while 'Risk' is checked. At the bottom, 'Cochran's and Mantel-Haenszel statistics' is unchecked, and the 'Test common odds ratio equals:' field is set to 1. There are buttons for '?', 'Cancel', and 'Continue' at the bottom.

☒ Chi-square

☐ Correlations

Nominal

☐ Contingency coefficient

☐ Phi and Cramer's V

☐ Lambda

☐ Uncertainty coefficient

Ordinal

☐ Gamma

☐ Somers' d

☐ Kendall's tau-b

☐ Kendall's tau-c

Nominal by Interval

☐ Eta

☐ Kappa

☒ Risk

☐ McNemar

☐ Cochran's and Mantel-Haenszel statistics

Test common odds ratio equals: 1

? Cancel Continue

FIGURE 7.4: Select the options

8

SPSS: inference about the difference between two means



Topics covered in this tutorial:

- Learn how to use SPSS for comparing two means.

8.1 Introduction

Two independent sample t -tests are used to compare the means of two separate groups.

Consider the face-plant data (Wojcik et al. 1999) used in Exercise 30.11 of the textbook¹ (Table 8.1).

TABLE 8.1: Lean-forward angles for older and younger women

Younger women		Older women
29	34	18
32	27	15
34	32	23
31	28	13
33	27	12



The data file (`forwardfall.sav`) can be downloaded in Sect. 10.

You can download the data, and follow the instructions.

8.2 Data entry

In the **Data View**, each row represents a unit of analysis (in this case, a person), as is usual in SPSS.

Each unit of analysis gets two variables recorded on it: one is quantitative (`LeanAngle`), and one is qualitative (`Group`). So there are two columns in the **Data View**, one for each variable.

The order doesn't matter, but here the quantitative variable is listed first (the lean-forward angle), and then the qualitative variable (the age group).

¹<https://bookdown.org/pkaldunn/SRM-Textbook/AnalysisTwoMeans.html>

The **Variable View** needs to be set up correctly too (Fig. 8.1). For example, all variables should be given a descriptive label. The quantitative variable (here, `LeanAngle`) should be declared as **Scale** in the **Measure** column. The qualitative variable (here, `Group`) should be declared as **Nominal** in the **Measure** column (Fig. 8.2).

 LeanAngle	 Group
29	1
34	1
33	1
27	1
28	1
32	1
31	1
34	1
32	1
27	1
18	2
15	2
23	2
13	2
12	2

FIGURE 8.1: The data in the Data View

	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure
1	LeanAngle	Numeric	8	0	Maximum lean...	None	None	11	 Right	 Scale
2	Group	Numeric	8	0		{1, Younger ...	None	8	 Right	 Nominal

FIGURE 8.2: The Variable View

The two age groups are defined so that a 1 represents younger women, and a 2 represents older women. Since this is arbitrary, we tell SPSS this by setting the **Values** in the **Variable View** (Fig. 8.3).

8.3 Analysis

1. **Use the SPSS menu:** select **Analyze> Compare Means> Independent Samples T Test...**
2. **Select the Variables.** The **Test Variable** is the quantitative variable (here, `LeanAngle`), and the **Grouping Variable** is the qualitative variable (here, ‘`Group`’); see Fig. 8.4.
3. **Define the groups to compare.** Click the button **Define Groups...** and enter the numbers that

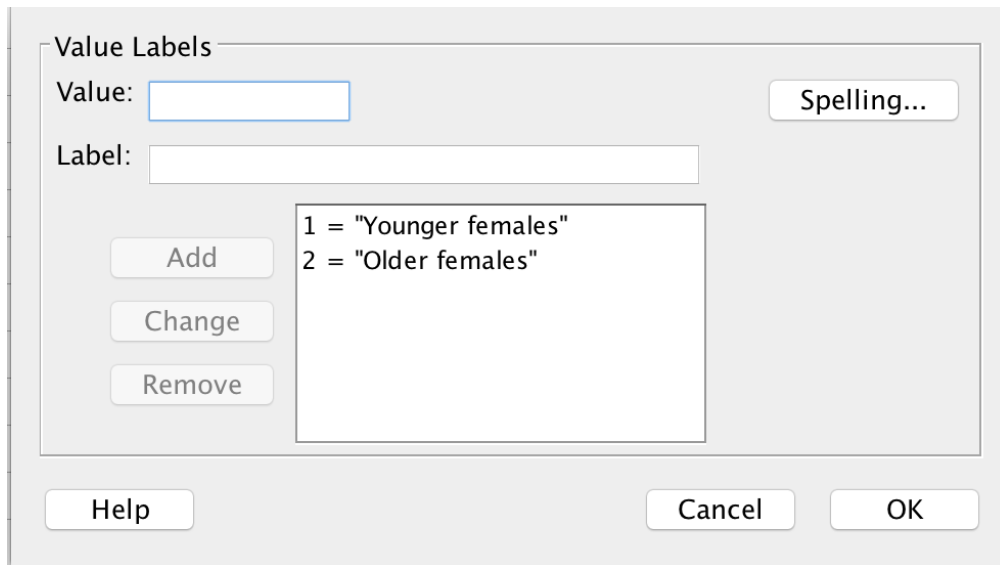


FIGURE 8.3: Defining the levels (Values) for the age group, in the Variable View

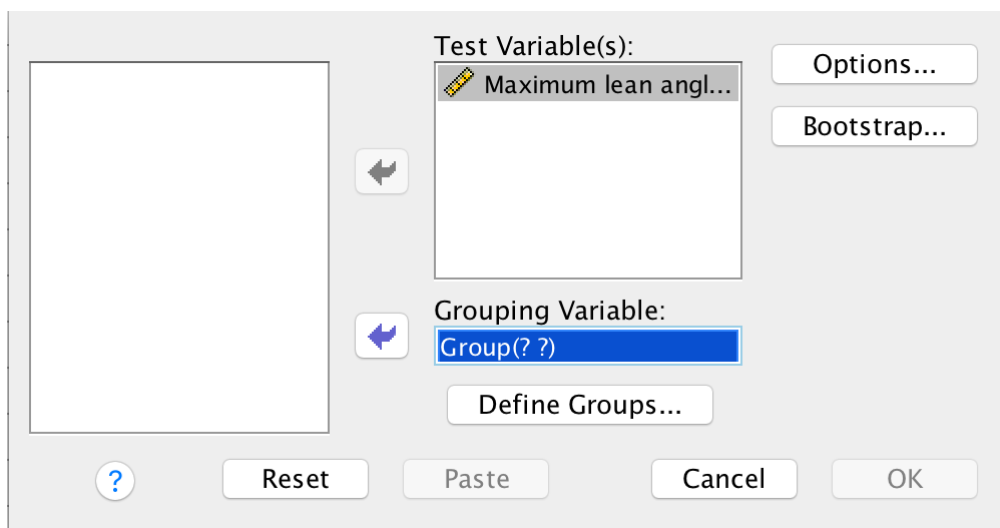


FIGURE 8.4: Entering the variables

are being used to represent the two groups to be compared. In this example, the two groups are given the values 1 and 2 in the **Data View**, so we enter these.

4. Click the **Continue** button, then **OK**. You will have your output in the SPSS Output window.

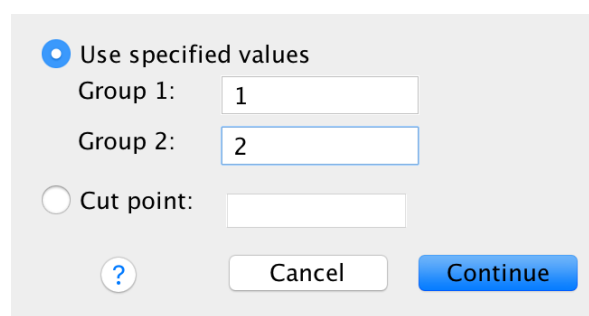


FIGURE 8.5: Telling SPSS which groups to compare

8.4 Numerical summaries

The *mean*, *median*, *standard deviation* and *standard error* are produced as part of the output from the *t*-test analysis step above.

The *mean* and *standard error* for the *difference* is obtained from the *t*-test output too; see Sect. 8.3.

9

SPSS: graphs



Topics covered in this tutorial:

- Creating graphs in SPSS.



The data files used in this section can be downloaded in Sect. 10.

You can download the data, and follow the instructions.

Graphs in SPSS are generally produced using the **Graphs** menu, and then selecting **Chart Builder...**

After making this selection, you usually get a warning (Fig. 9.1)). It is reminding you that it is **really important** that you have declared the **Measure** correctly for each variable in the **Variable View** (that is, correctly declared each variable as **Nominal**, **Ordinal** or **Scale**); see Sect. 6.

Provided you have done so, you can press **OK**.

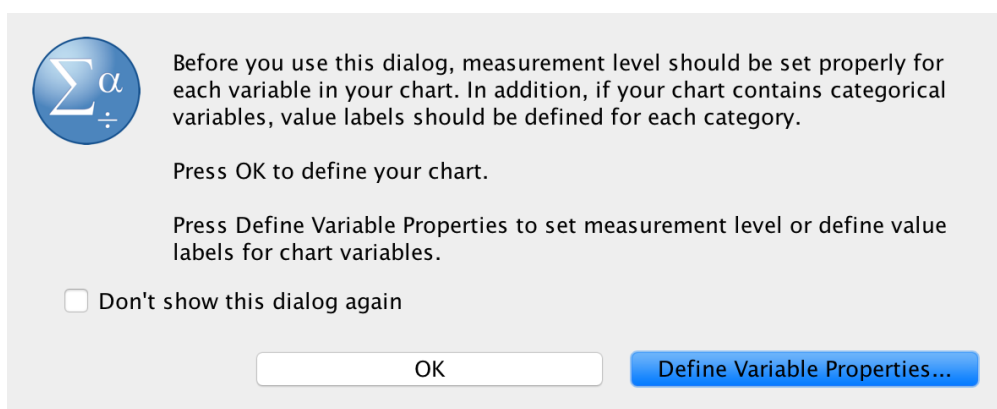


FIGURE 9.1: The warning after selecting *Graphs*



The graphs produced by default in SPSS are usually an inconvenient size for copying-and-pasting into Word, PowerPoint, and other places. We now explain the easy solution.

When graphs are created in SPSS using the **Chart Builder**, the window shown in Fig. 9.2 initially appears. Select the **Options** item from the right-hand side of this window.

Then, change the **Chart size** to (say) 60% or a similar value (Fig. 9.3). Now, when you copy-and-paste the graph, the sizing will be more appropriate.

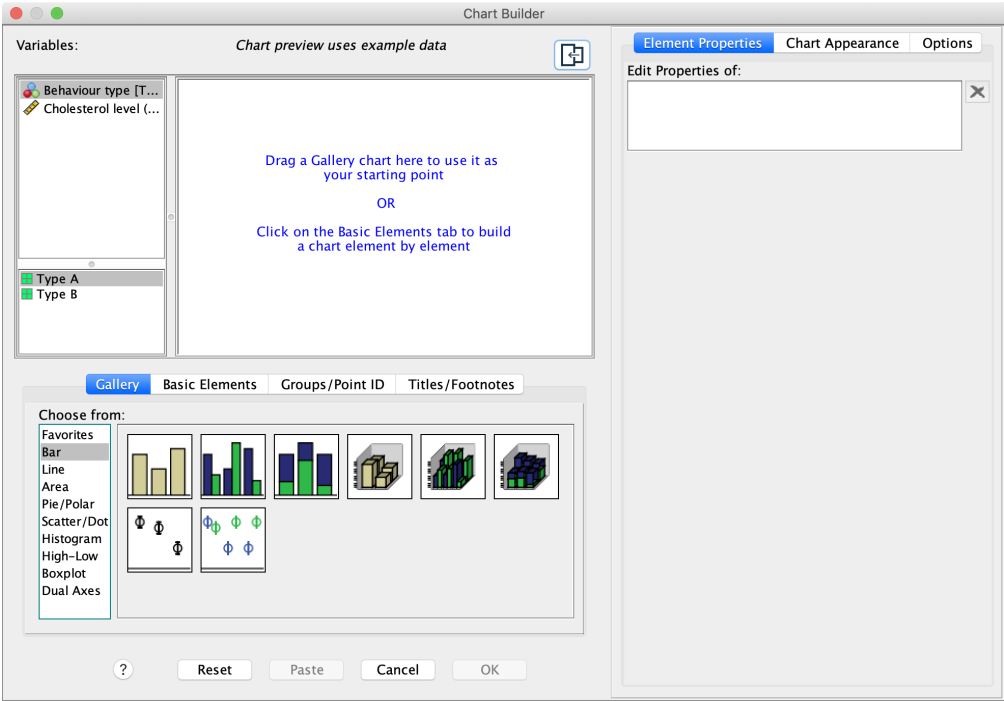


FIGURE 9.2: The Chart Builder window

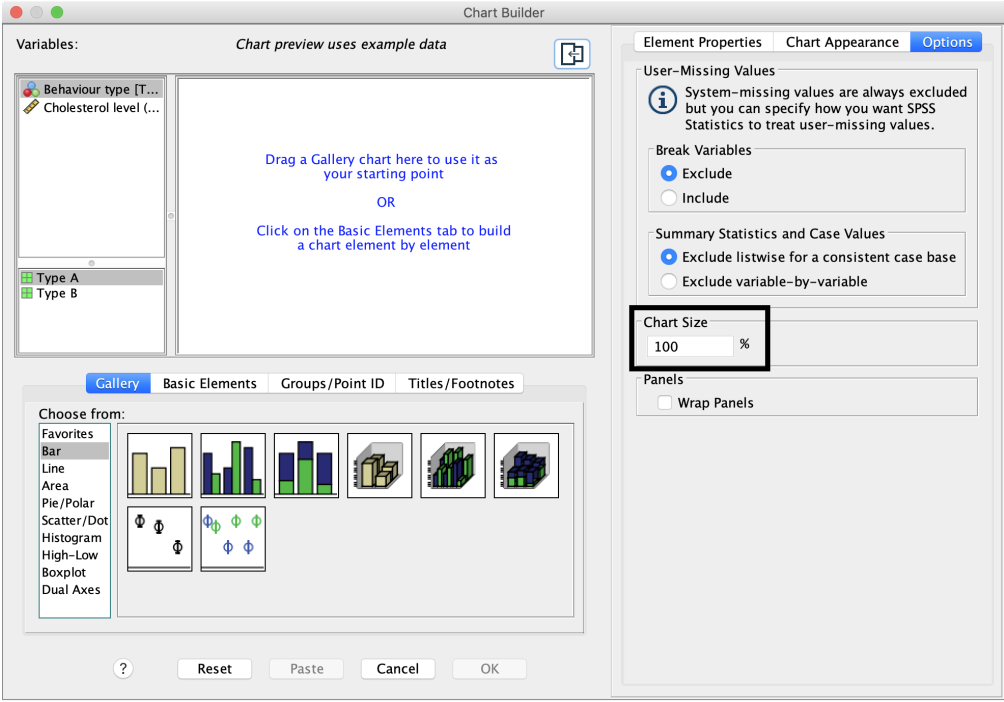


FIGURE 9.3: The warning after selecting Graphs

9.1 Side-by-side bar charts

For this section, the data introduced in Sect. 7.1 are used.

1. **Use the SPSS menu:** select **Graphs> Chart Builder...**
2. **Select the type of plot.** Select **Bar** from the Gallery, then select the **second** template from the top row, and drag it to the canvas.

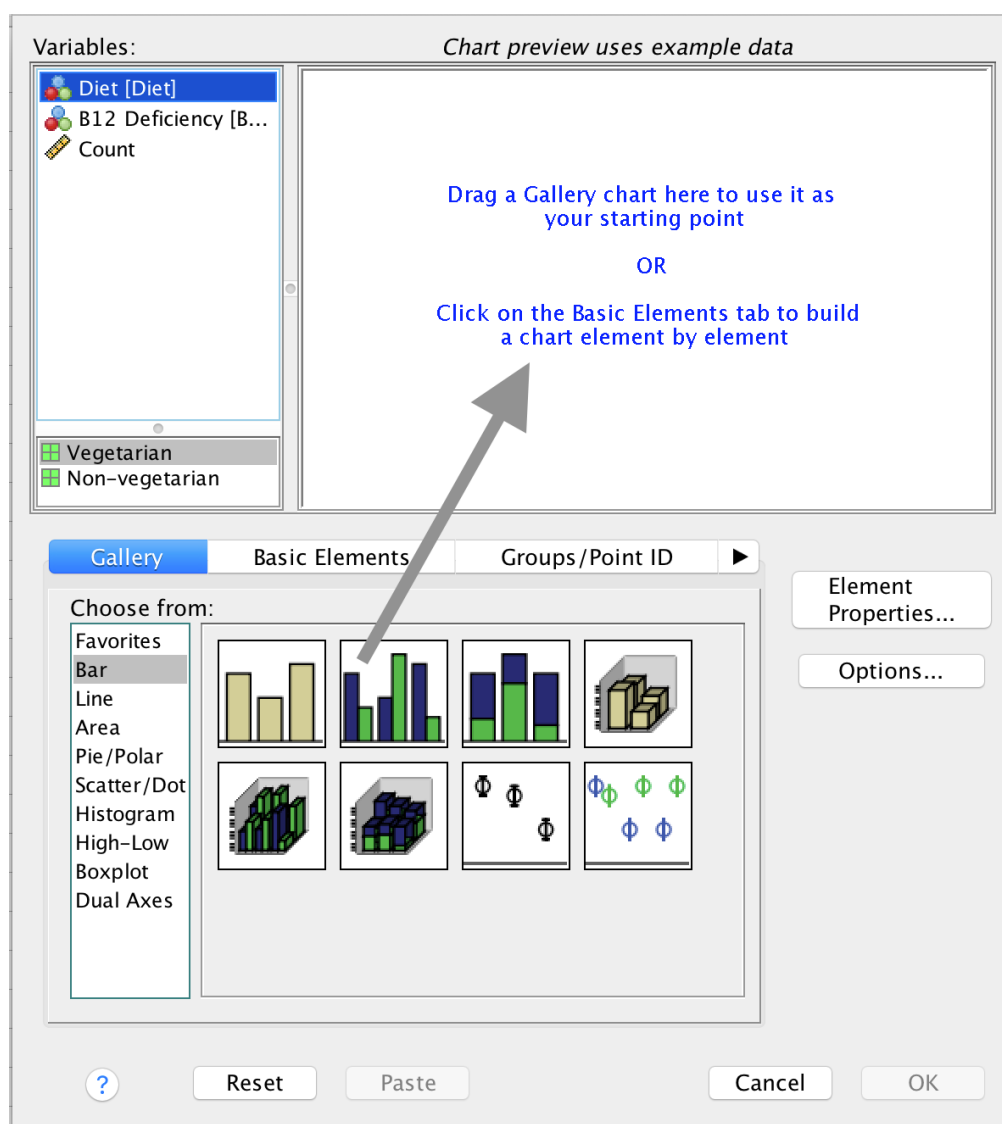


FIGURE 9.4: Selecting a side-by-side bar chart

3. **Add the variables.** Drag one qualitative variable to the bottom (left-to-right) axis; this can often be seen as the explanatory variable. Drag the other to the top right (where it says, cryptically, **Cluster on X: set color**); see Fig. 9.5).
4. Press **OK**. You will have your graph in the SPSS Output window.

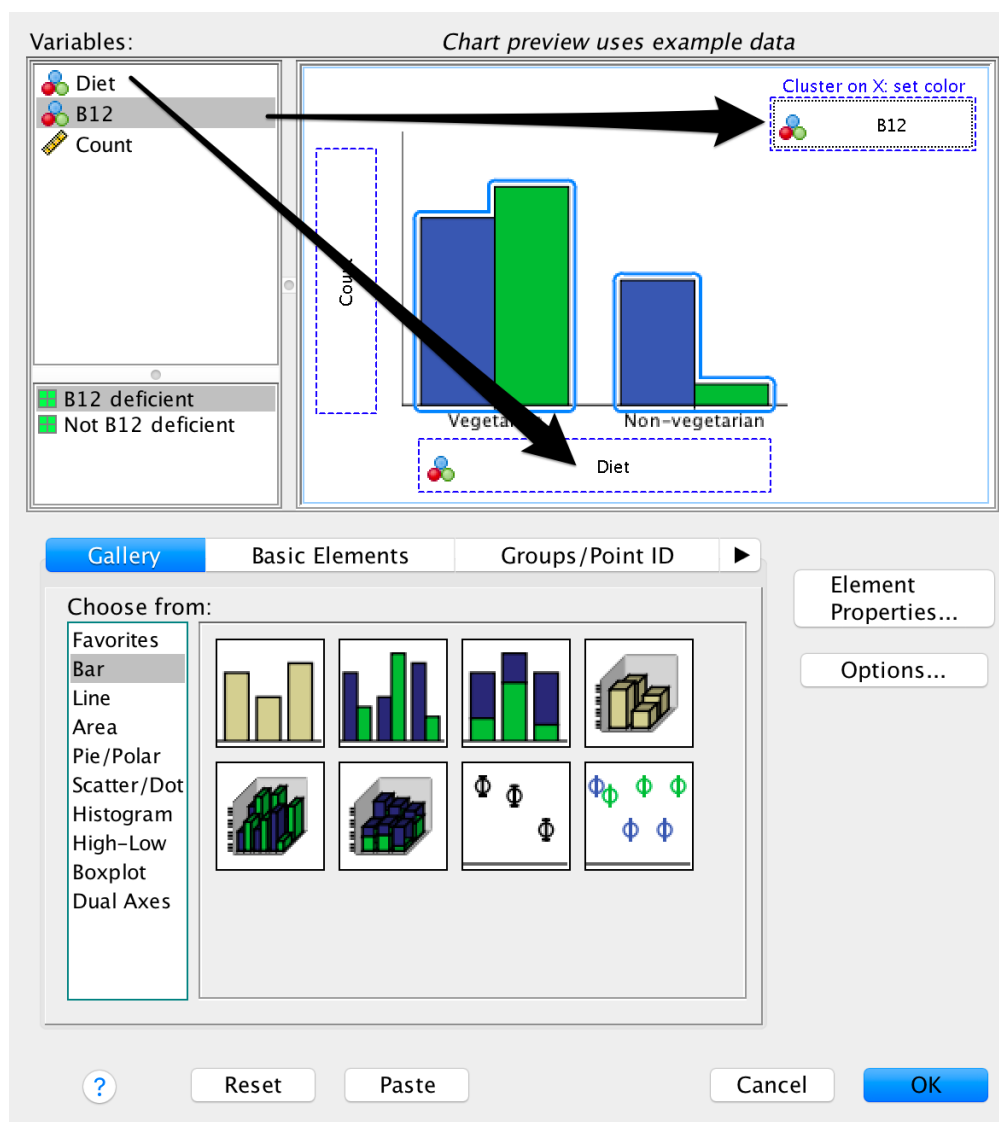


FIGURE 9.5: Adding the variables to create a side-by-side bar chart

9.2 Stacked bar charts

For this section, the data introduced in Sect. 7.1 are used.

1. **Use the SPSS menu:** select **Graphs> Chart Builder...**
2. **Select the type of plot.** Select **Bar**, then select the **third** template from the top row, and drag it to the canvas (Fig. 9.6).
3. **Add the variables.** Drag one qualitative variable to the bottom (left-to-right) axis; this can often be seen as the explanatory variable. Drag the other to the top right (where it says, cryptically, **Stack: set color**); Fig. 9.7.
4. Press **OK**. You will have your graph in the SPSS Output window.

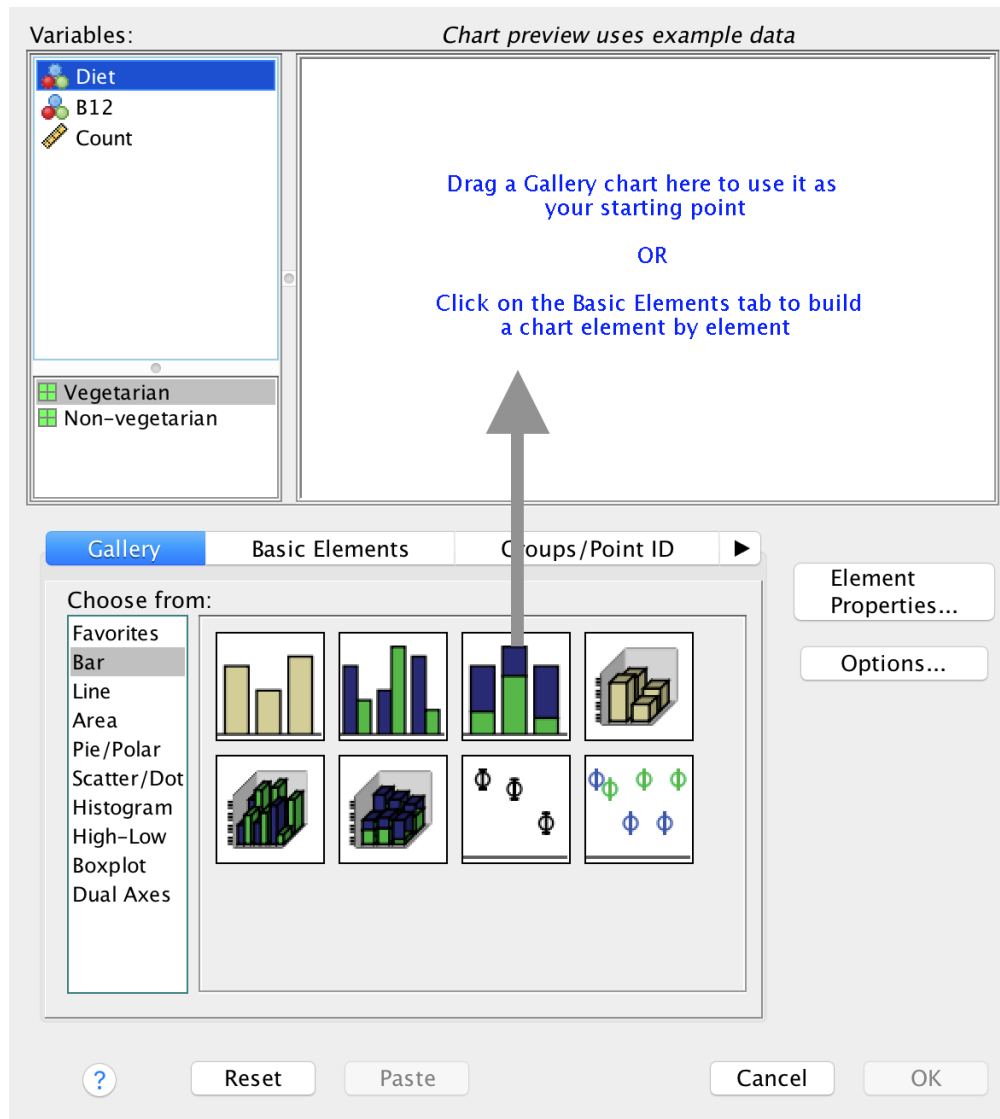


FIGURE 9.6: Selecting a stacked bar chart

9.3 Boxplots

For this section, the data introduced in Sect. 8.1 are used.

1. **Use the SPSS menu:** select **Graphs> Chart Builder...**
2. **Select the type of plot.** Select **Boxplot**, then select the **first** template from the top row, and drag it to the canvas (Fig. 9.8).
3. **Add the variables.** Drag the qualitative variable to the bottom (left-to-right) axis; drag the quantitative variable to the vertical (up-and-down) axis (Fig. 9.9).
4. Press **OK**. You will have your graph in the SPSS Output window.

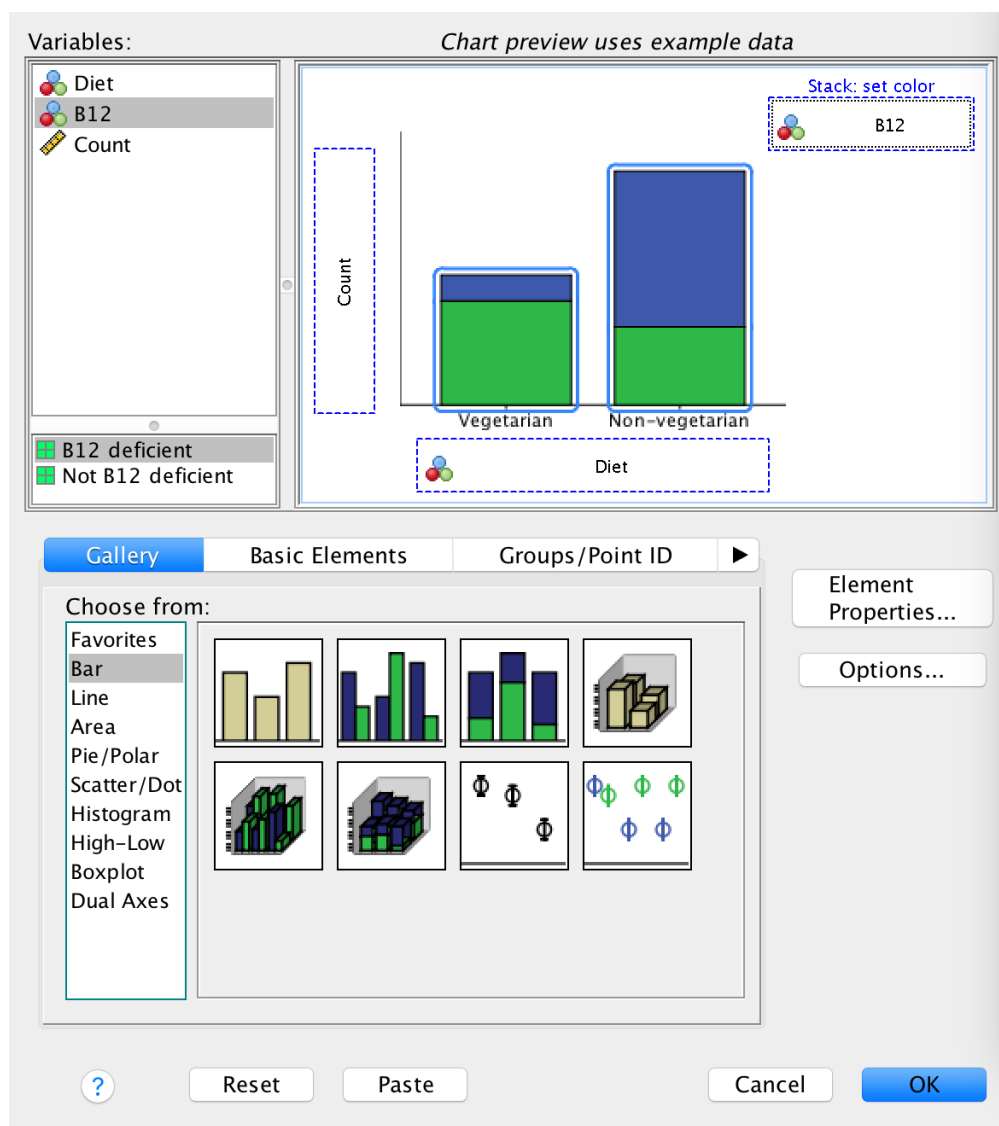


FIGURE 9.7: Adding the variables to create a stacked bar chart

9.4 Error bar charts

For this section, the data introduced in Sect. 8.1 are used.

1. **Use the SPSS menu:** select **Graphs> Chart Builder...**
2. **Select the type of plot.** Select **Bar**, then select the **third** template from the **second row**, and drag it to the canvas (Fig. 9.10).
3. **Add the variables.** Drag the qualitative variable to the bottom (left-to-right) axis; drag the quantitative variable to the vertical (up-and-down) axis (Fig. 9.11).
4. Press **OK**. You will have your graph in the SPSS Output window.

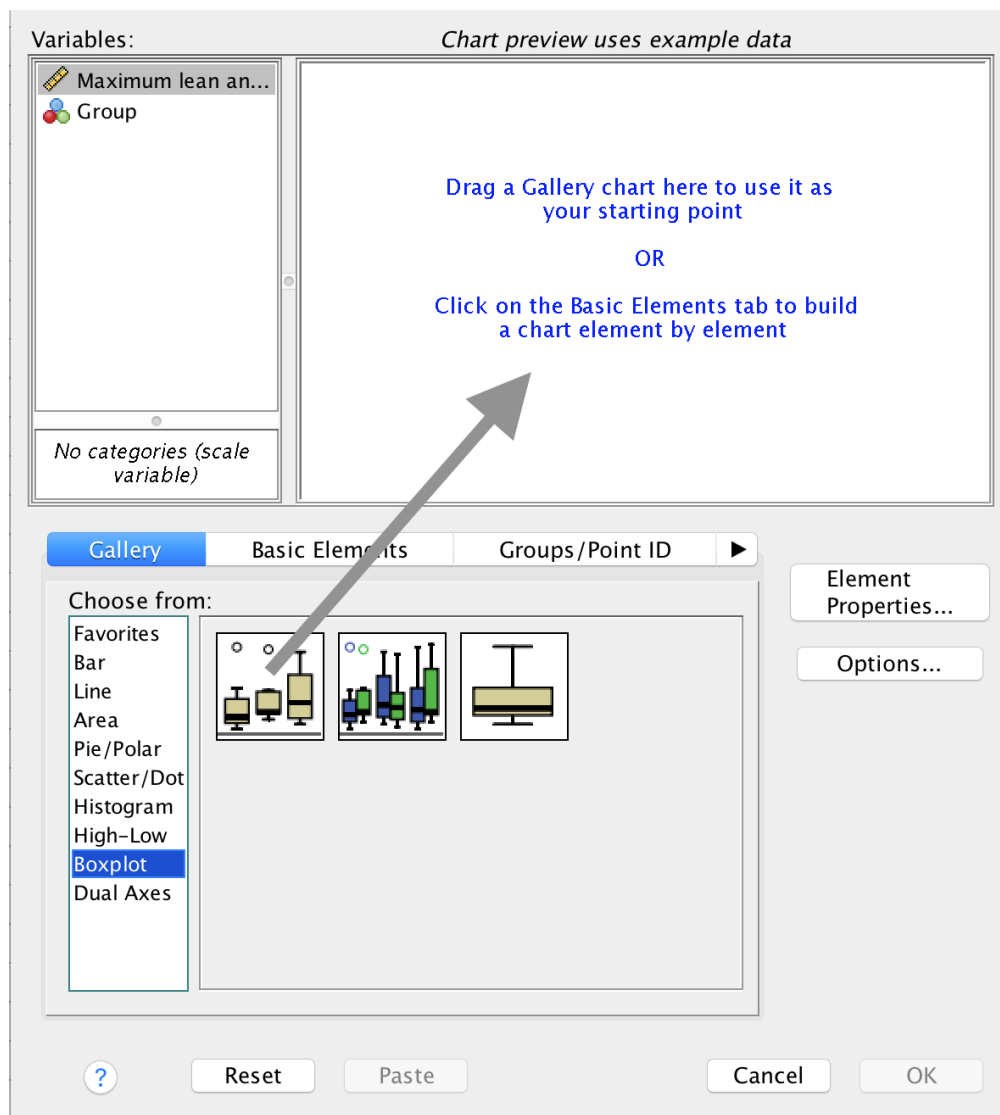


FIGURE 9.8: *Selecting a boxplot*

9.5 Editing graphs

Your graph appears in the SPSS Output window. Double-clicking on the graph opens the **Chart Editor**.

In the **Chart Editor**, you can edit almost any part of the graph, by clicking on various elements of the graph (e.g. the axis labels) or clicking on the icons at the top of the Chart Editor window.

Once you have finished editing your graph in the **Chart Editor**, close that window, and the graph will be updated in the SPSS Output Window.

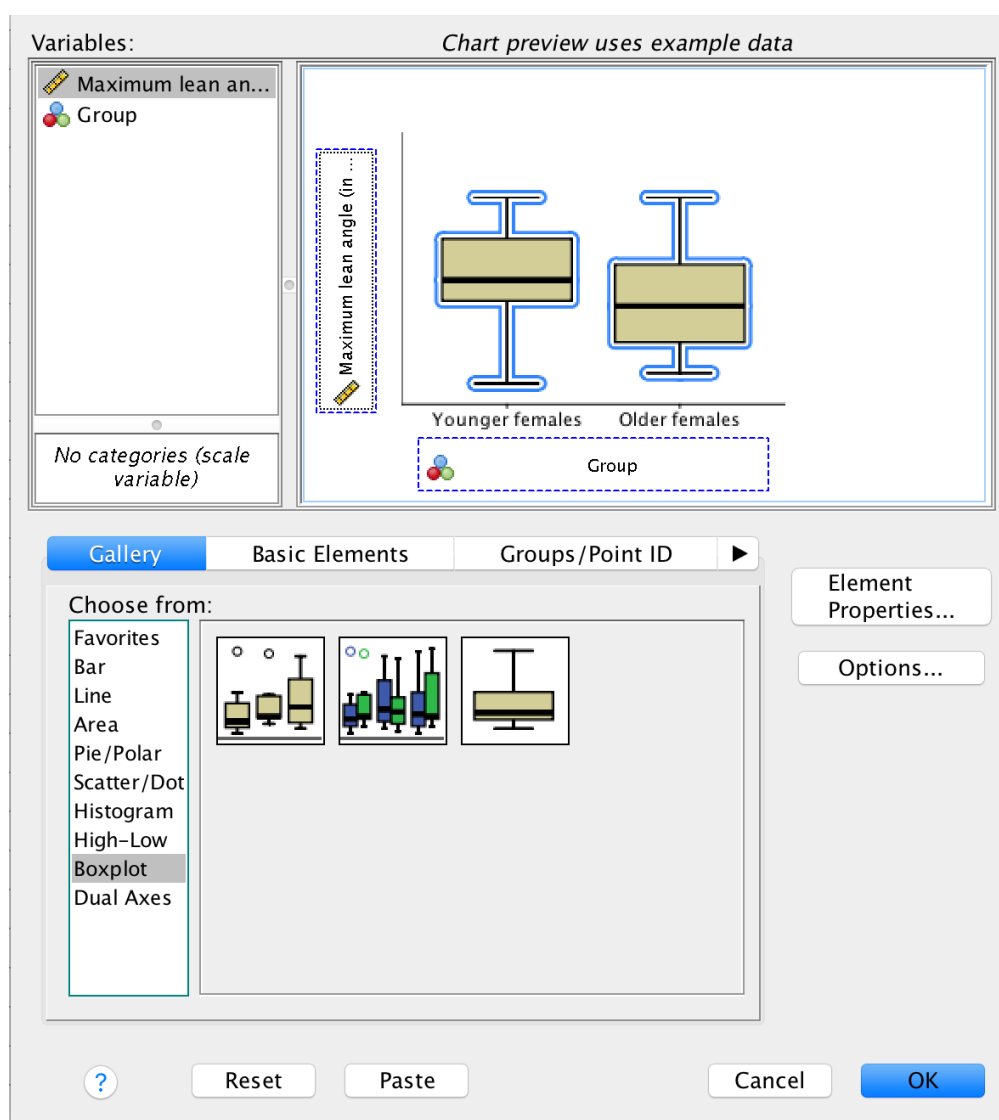


FIGURE 9.9: Adding the variables to create a boxplot

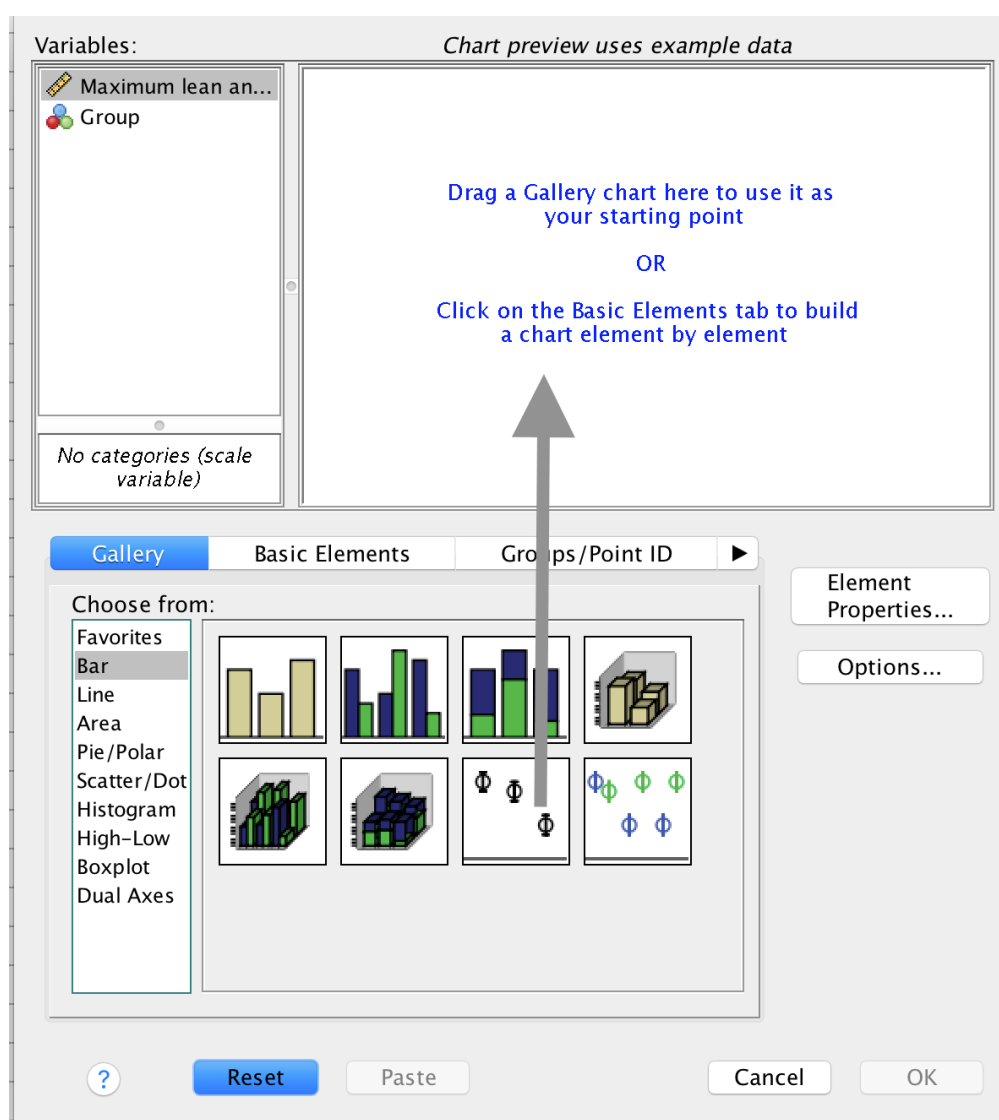


FIGURE 9.10: Selecting an error bar chart

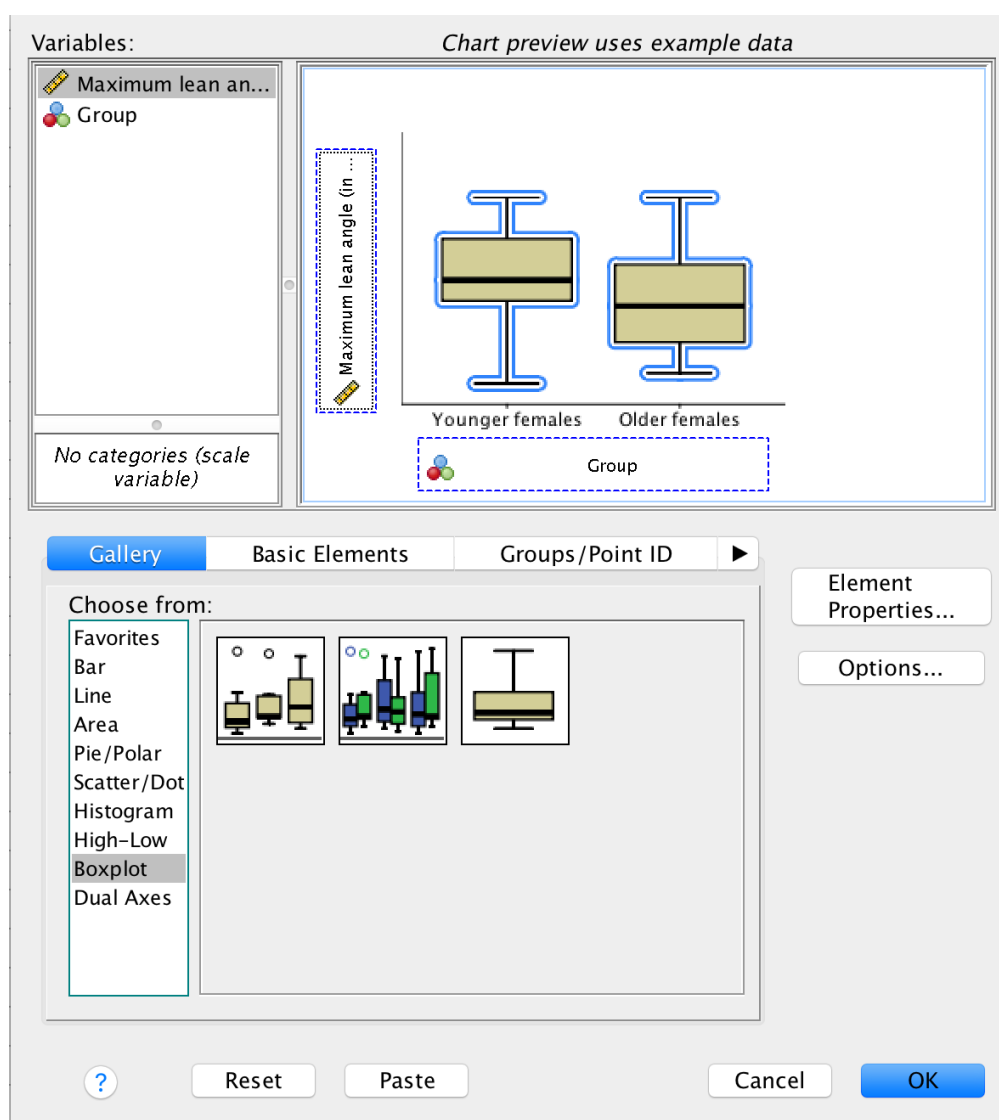


FIGURE 9.11: Adding the variables to create an error bar chart

10

Data files

The following data sets are available for download.

Data	Software	Reference	Download
B12 data	jamovi	Sect. 3	B12-Long.csv ¹
B12 data	SPSS	Sect. 7	B12-Long.sav ²
Face-plant data	jamovi	Sect. 4	forwardfall.csv ³
Face-plant data	SPSS	Sect. 8	forwardfall.sav ⁴

¹ [./Data/B12-Long.csv](#)

² [./Data/B12-Long.sav](#)

³ [./Data/forwardfall.csv](#)

⁴ [./Data/forwardfall.sav](#)

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